



Figure 1: Ceph architecture

which stores the data, checks its own state and other OSDs and reports back to monitors. In case an OSD goes down, the Ceph OSD Daemon automatically triggers data rebalancing based on the new cluster map from Ceph Monitor. Data is always transferred from the OSD nodes and never from the monitors to the clients.

This architecture of Ceph might seem familiar to those who know about Google File System (GFS) and Hadoop Distributed File System (HDFS) but it is also very different from them in multiple ways. Ceph uses an algorithm known as CRUSH (controlled, scalable, decentralised placement of replicated data) for random and distributed data storage among the OSDs. Ceph doesn't need two round-trips for data retrieval like HDFS or GFS, in which one trip is to the central lookup table to find the data location and the second trip is to the located data node. Every bit of data stored in Ceph OSDs is self-calculated using the CRUSH algorithm and stored independent of any other attribute. When a client requests data from Ceph, this CRUSH algorithm is used to find the exact location of all the requested blocks, and the data is transferred by the responsible OSD nodes.

As and when any OSD goes down, a new cluster map is generated and the duplicate data of the crashed OSD is transferred to a new node based on results from the CRUSH algorithm.

Ceph as object storage

Object stores have certain advantages over traditional block storage. Every stored object is given an object ID, which makes it faster to search/access and easier to manage. An object store is highly scalable for a large number of objects but has limitations on the size of a single object.

Ceph provides the LibRADOS native library, which allows direct access to RADOS using apps with support for C, C++, Java, Python, Ruby and PHP. Ceph also features RADOSGW,

a RESTful HTTP API as the object gateway, which is compatible with Amazon's S3 and Openstack's Swift services. Ceph can be used to make a cloud object storage solution like Dropbox or Google Drive using the Openstack cloud platform.

HDFS, GFS, Gluster-Swift, EMC Atmos and NetApp StorageGRID Webscale are a few of the leading proprietary object store systems.

Ceph as block storage

Block store is the traditional form of disk data storage where the data is divided into blocks and stored using a file system. Block store is best suited for VM disk volume storage needs, where we store large singular files with higher read and write frequencies. Ceph block storage can be well served as a SAN (storage area network) solution.

One added advantage of Ceph is that it has thin-provisioning enabled, which helps in faster data duplication and efficient storage space utilisation. When a disk stored in Ceph is replicated, it doesn't occupy any memory for the replicated copy. Ceph uses the COW (copy-on-write) method where the block is replicated and updated only on write, else the block is not replicated.

Amazon's EBS (Elastic Block Storage) is a popular block storage service used generally for volumes of VM instances.

Ceph as file system storage

A file system based storage is like any NAS (network attached storage) system, where the file system is managed by a remote storage device. Ceph uses the Ceph FS (Ceph file system), which provides a POSIX-compliant file system as an interface. A client system can mount this file system and access the file storage.

Ceph FS uses Ceph MDS (Ceph metadata server) in the storage cluster to store the metadata about the file system. This metadata consists of the file system tree structure, timestamps, permissions and other POSIX-compliant data. Ceph MDS can also be distributed by dividing the tree structure and storing it in different Ceph MDSs. This metadata is stored like any other object in the Ceph OSDs, and is replicated for backup and recovery purposes. FreeNAS, NAS4Free are some of the NAS solutions available. The Lustre file system is a famous open source parallel file storage solution. Oracle ZFS Storage and Dell Storage SCv2000 are a few of the popular NAS solutions from proprietary vendors.

Ceph in cloud platforms

Storage is an important component in the cloud service providers' (CSP) list of services. Major CSPs provide multiple forms of cloud storage services, which include rapid storage, archival storage, object store, block store, NAS, SAN, VM disk storage, etc. All these solutions generally require different dedicated hardware and software in order to